Operating Systems
Week 2
Lecture/Discussion 2

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Previous

- Basic features (Concurrence, Sharing, Virtual)
- Operating System services

Today Discussion

- How OS services can be managed
- Examples of Real-Time systems

The Role of an OS

A computer is a set of resources for the movement, storage, and processing of data.

The OS is responsible for managing these resources.

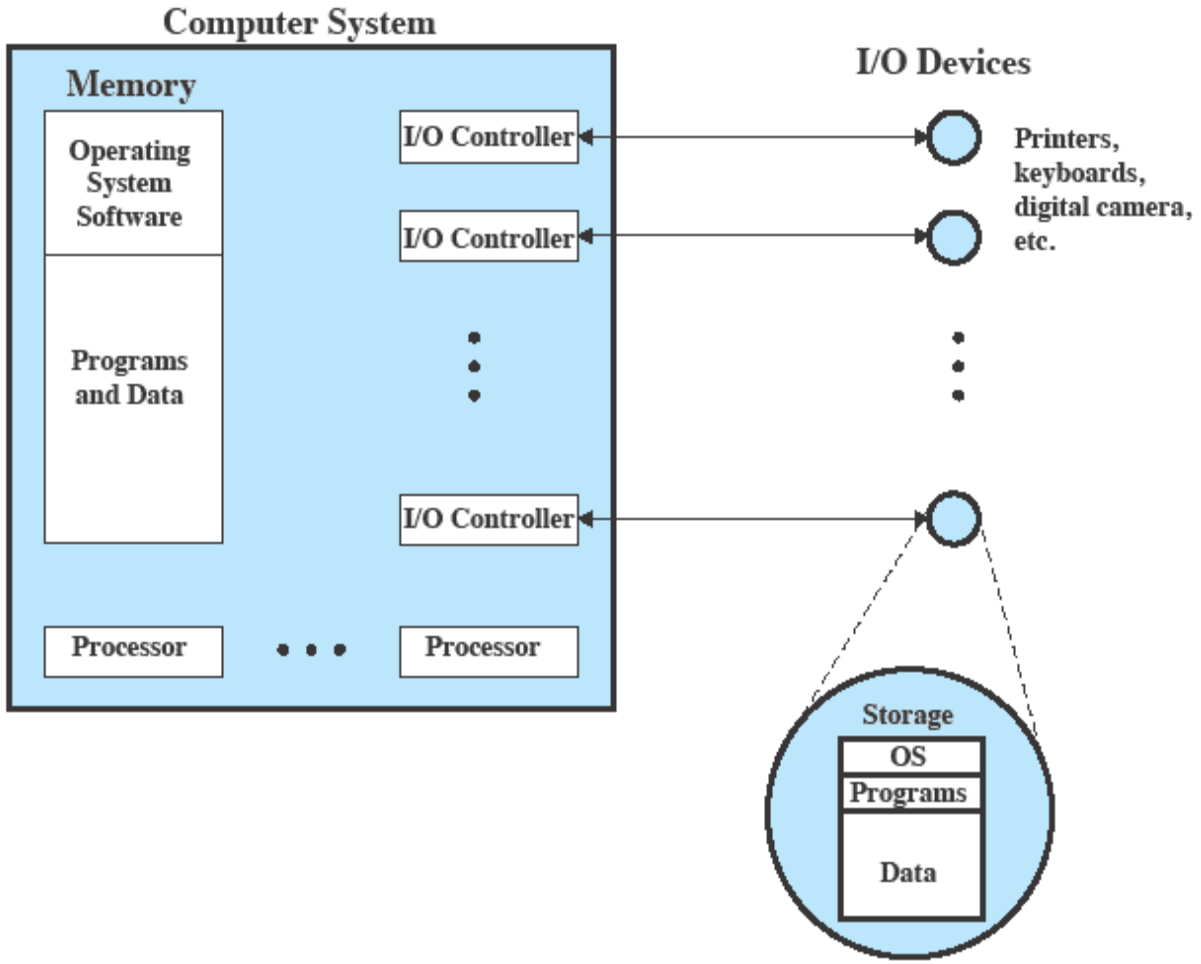
Operating System as Software

The OS functions in the same way as an ordinary computer software

- It is a program that is executed by the CPU

Operating system relinquishes control of the processor

OS as Resource Manager



Evolution of Operating Systems

Operating systems will evolve over time

- Hardware upgrades plus new types of hardware
- New services
- Fixes

Process Description and Control

How are processes represented and controlled by the OS.

Process states which characterize the behaviour of processes.
Data structures used to manage processes.
Ways in which the OS uses these data structures to control process execution.
Discuss process management in UNIX SVR4.

Requirements of an Operating System

Fundamental Task: Process Management

The Operating System must

- Interleave the execution of multiple processes
- Allocate resources to processes, and protect the resources of each process from other processes,
- Enable processes to share and exchange information,
- Enable synchronization among processes.

Concepts

From earlier topics we saw:

Computer platforms consists of a collection of hardware resources

Computer applications are developed to perform some task

It is inefficient for applications to be written directly for a given hardware platform

Concepts cont…

OS provides an interface for applications to use

OS provides a representation of resources that can be requested and accessed by application

The OS Manages Execution of Applications

Resources are made available to multiple applications
The processor is switched among multiple application
The processor and I/O devices can be used efficiently

What is a “*process*”?

A program in execution
An instance of a program running on a computer
The entity that can be assigned to and executed on a processor
A unit of activity characterized by the execution of a sequence of instructions, a current state, and an associated set of system instructions

Process Elements

A process is comprised of:

- Program code (possibly shared)
- A set of data
- A number of attributes describing the state of the process

Process Elements

While the process is running it has a number of elements including

- Identifier
- State
- Priority
- Program counter
- Memory pointers
- Context data
- I/O status information
- Accounting information

Process Control Block

Contains the process elements
Created and manage by the operating system
Allows support for multiple processes

